AGODA · SUPPLY ANALYTICS CASE STUDY

THE URGENCY MESSAGE

Do hotel prices really rise as check-in approaches & where should an urgency nudge be?

49,061
Bookings Analyzed

5 cities
Q4 | Oct-Dec 2016, 880 hotels

~ 0
Real price trend to check-in

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EXECUTIVE SUMMARY

The answer, before the journey

What I found, what it means, and what to do about it.

Bottom line: Hotel prices do not systematically rise as the check-in date approaches, the apparent 39.5% move is selection bias. So, urgency messaging should be honest and targeted : price-rise only where the data backs it.

What I tested

Across 49K bookings, how the price paid moves as the check-in date approaches.

What's true

Within the same room-night the trend is ~0 (p=0.88). Prices are volatile, not directional.

The opportunity

City D + Weekend + 4-5 star rises 82% of the time - a genuine, honest target

What to do

Reserve price-rise for proven segments; A/B test before scaling.

From a tempting trend to a causal answer

The Product Owner wants an urgency message. First, I had to check if its premise is even true.

The premise to test

"Prices have been rising, book now to lock in your rate."

If true, urgency messaging is justified. If not, I'd be misleading customers - a trust and compliance risk. The data must decide.

Step 1 **Explore**

Profile 49064 bookings; map how price behaves as the check-in date approaches, fix incorrect accommodation type name & **remove** check-in date < booking date cases (3 cases).

Step 2 **Control for mix**

Hold star rating, chain, city & weekend constant, then compare each hotel to itself : strips out which hotels book when, not just which prices move.

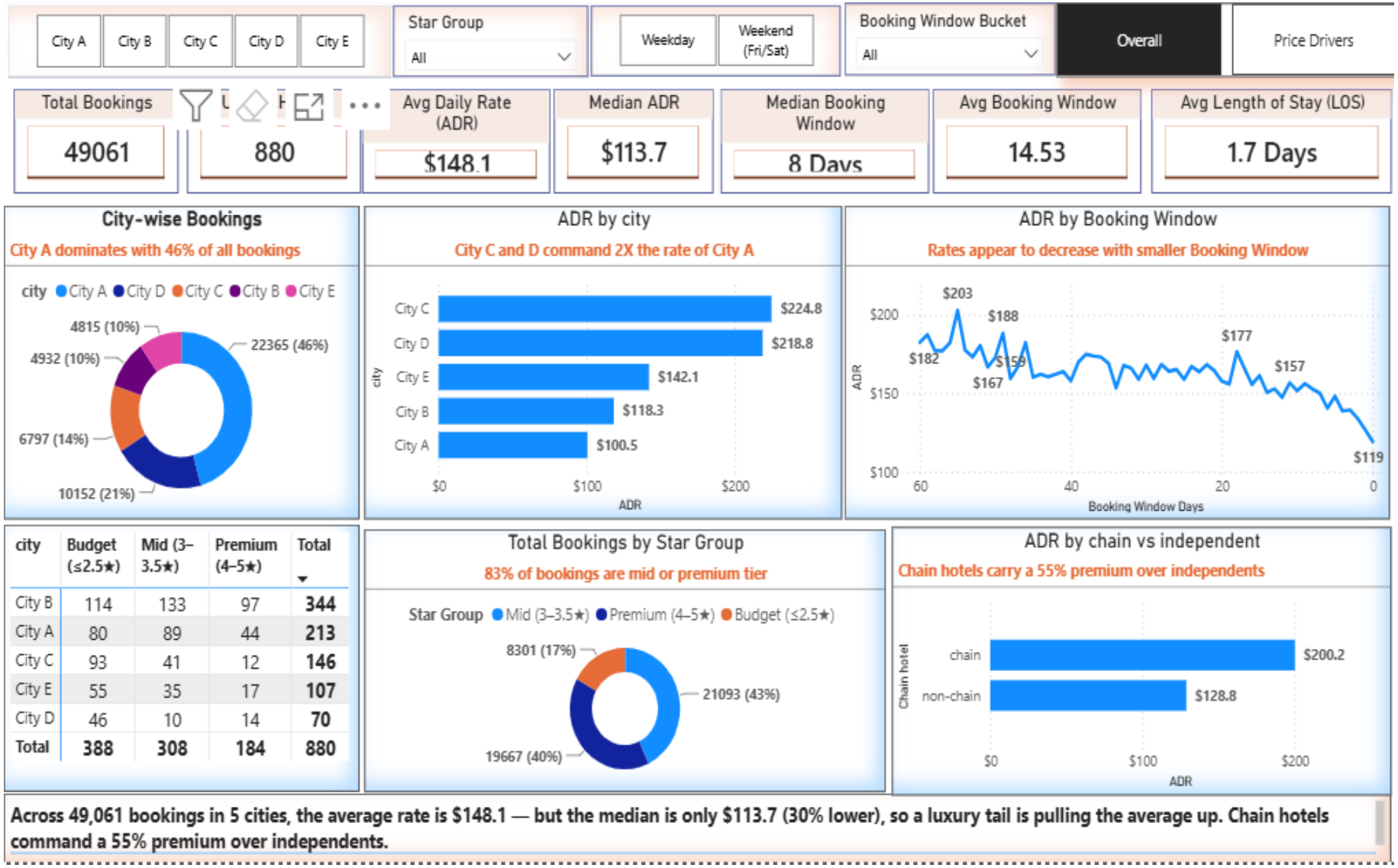
Step 3 **Isolate the signal**

Narrow to the same hotel on the same night, booked at different times : the clean test, where only timing varies.

Step 4 **Translate**

Turn the finding into a targeted, honest recommendation where the message is true, where it misleads, and what to do.

Live & Interactive Dashboard : Executive Overview



city	Budget (≤2.5★)	Mid (3–3.5★)	Premium (4–5★)	Total
City B	114	133	97	344
City A	80	89	44	213
City C	93	41	12	146
City E	55	35	17	107
City D	46	10	14	70
Total	388	308	184	880

Across 49,061 bookings in 5 cities, the average rate is $148.1 — but the median is only $113.7 (30% lower), so a luxury tail is pulling the average up. Chain hotels command a 55% premium over independents.

A clean, well-behaved sample : with one telling detail in the averages.

- **Median over mean :** $148 avg vs $114 median - a luxury tail inflates the average.
- **High intent, late funnel : 47.9%** book within 8 days of check-in date
- **Length of Stay : 1.7 Days** is the average stay length of the bookings
- **Late-funnel :** Half of all bookings happen within 8 days of check-in.
- **Concentrated :** City C & D command ~2x the rate of City A; chain hotels carry a 55% premium.

The tempting first look : prices appear to drop toward check-in

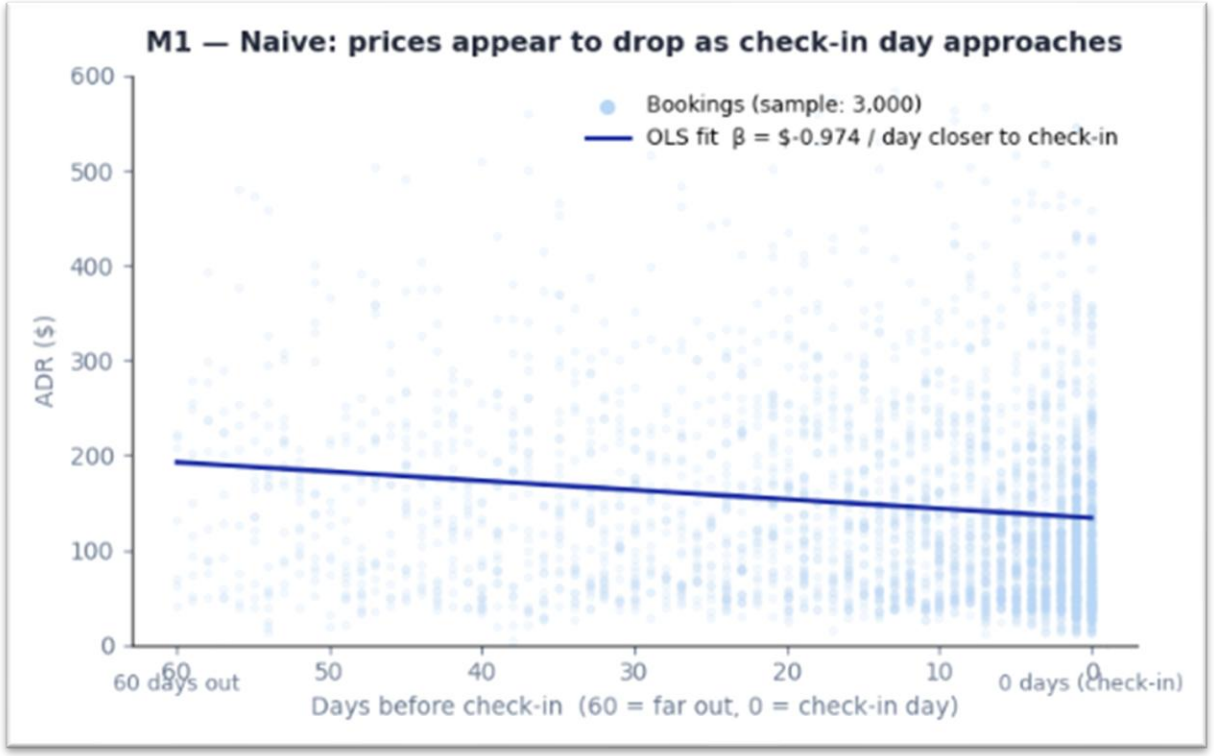
ADR appears ~39.5% higher far out and lower toward check-in : the apparent case for urgency.

-$0.974 / day × 60 days = –$58 swing (–39.5%)

Implied ADR move across 60 to 0 days before check-in

The trap: This compares different bookings for different hotels. Reading 60 days out down to check-in, the raw line slopes down, so prices look **lower** near check-in : so "prices are rising" is already wrong for the average guest.

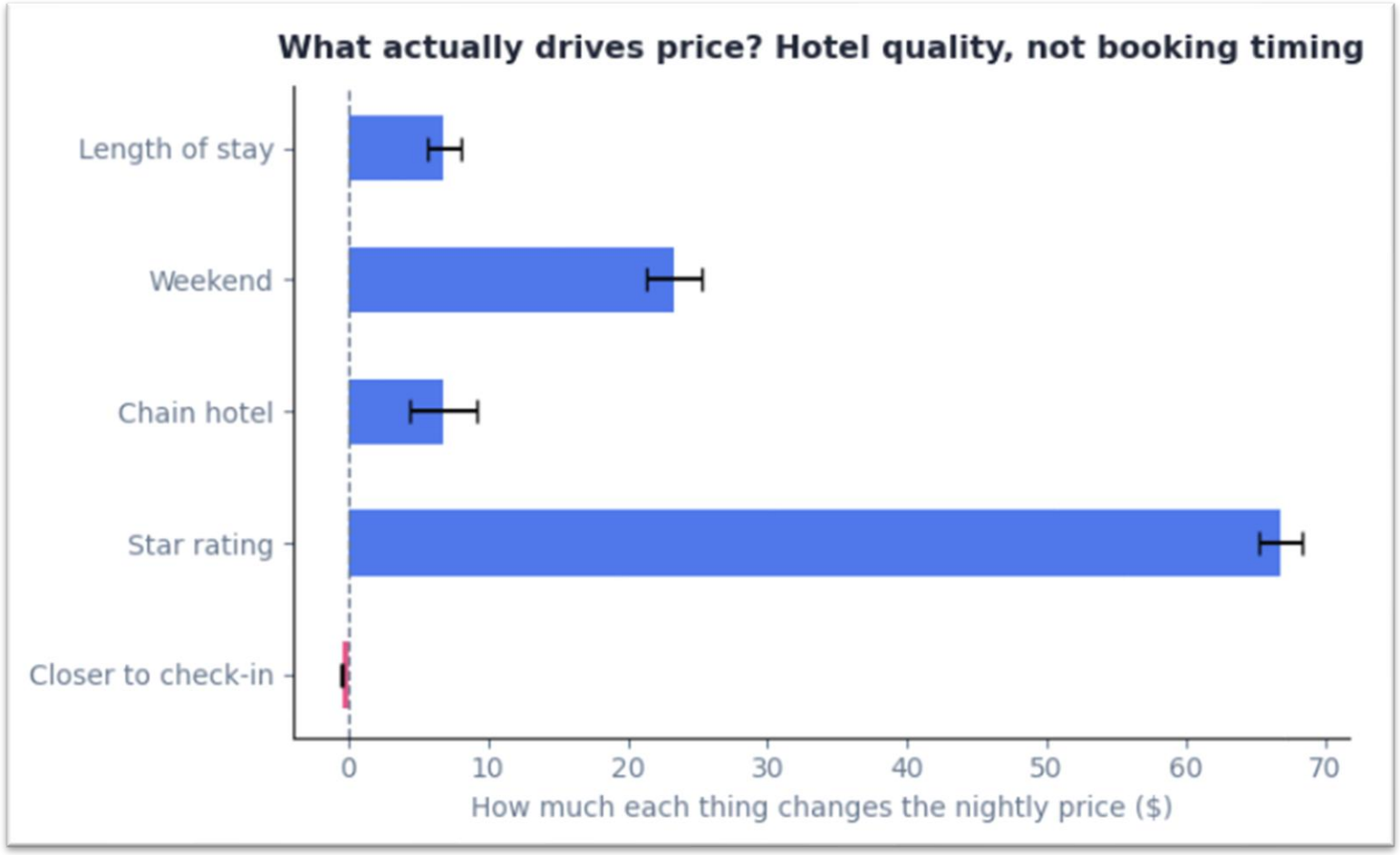
M1 : ADR vs booking window (every booking)



$R^2 = 0.013$. Booking Window explains barely 1% of price. Something structural is doing the real work.

Model 2 : Hold hotel quality and season constant

Add star rating, chain, weekend, length of stay, city and month. The timing effect shrinks toward zero (-17%).



-$0.974 → -$0.423 / day

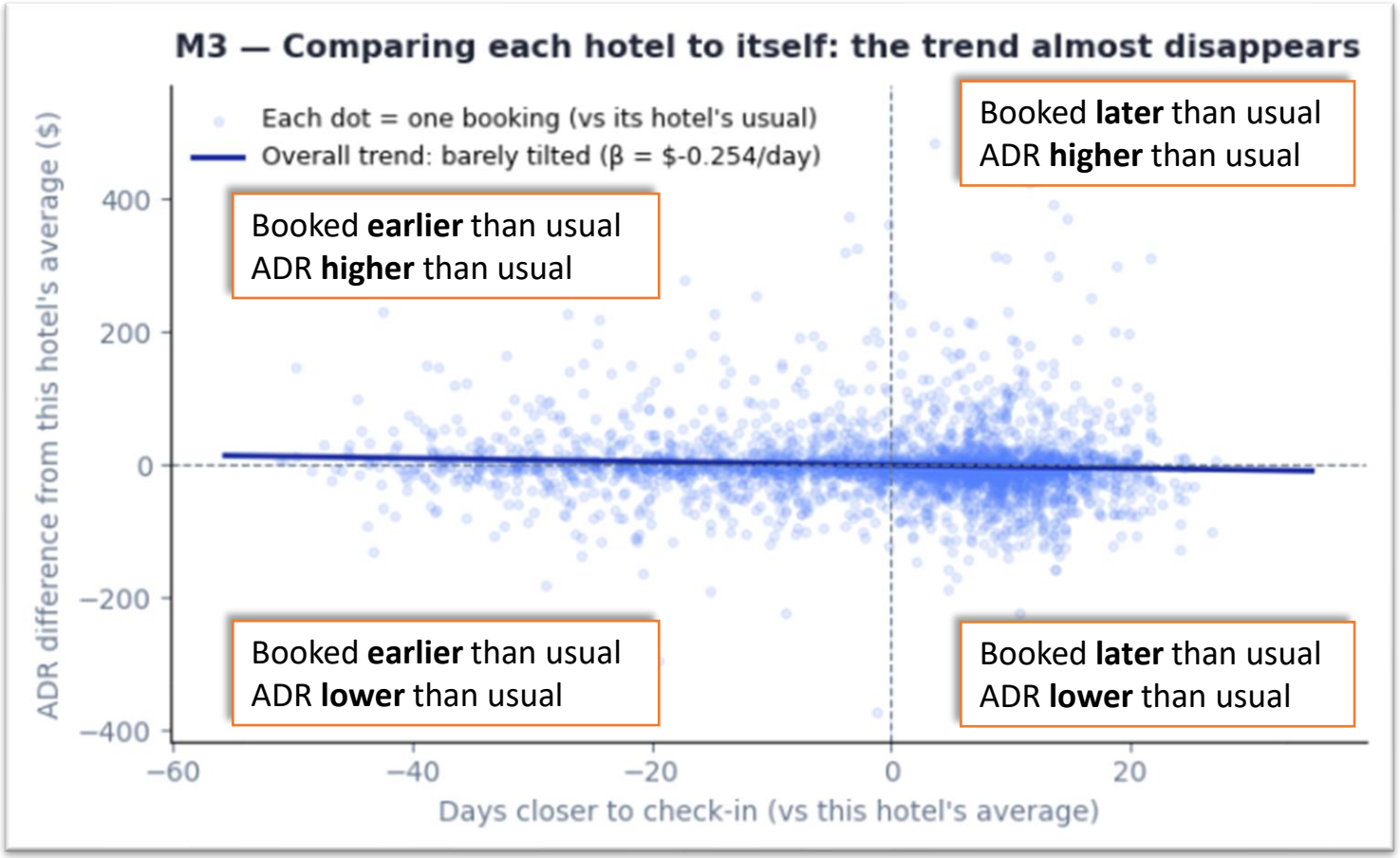
the per-day effect (closer to check-in) after observable controls

- **Star rating dominates.** A star upgrade is worth ~$67, it dwarfs every other driver and is exactly what M1 ignored.
- **Timing is tiny.** The "closer to check-in" bar is barely visible next to quality : when you book matters far less than what the hotel is.
- **R^2 jumps to 0.41.** Quality, city and season explain 41% of price, versus 1% for timing alone.

Still mixes different hotels together : M3 compares each hotel only to itself.

Model 3 : Compare each hotel to itself

Subtract each hotel's own averages : only within-hotel variation remains. The effect shrinks further (-10%).



-$0.423 → -$0.254 / day

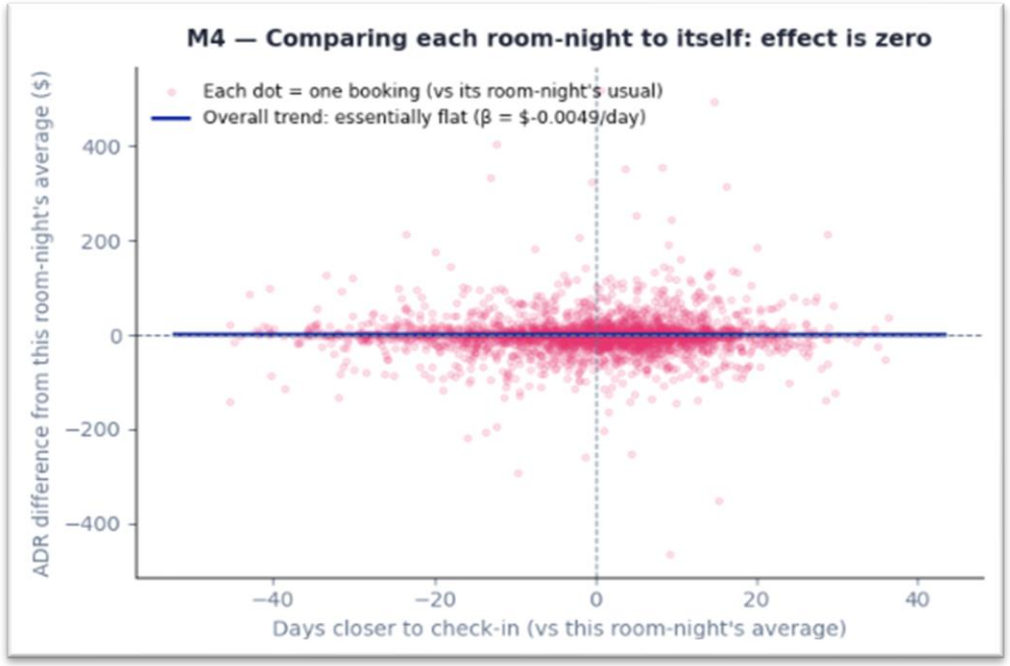
within-hotel effect, clustered by hotel

- **The mix is gone.** Each hotel is compared only to itself, so hotel-quality bias is removed by construction.
- **The line goes flat.** Across a -$400 to +$500 spread the slope barely tilts - timing is nearly irrelevant within a hotel.
- **A third still vanished.** The shrink from -$0.42 to -$0.25 was unobserved quality that star rating alone could not capture.

Within a hotel, room types still differ : M4 compares the same room-night to itself.

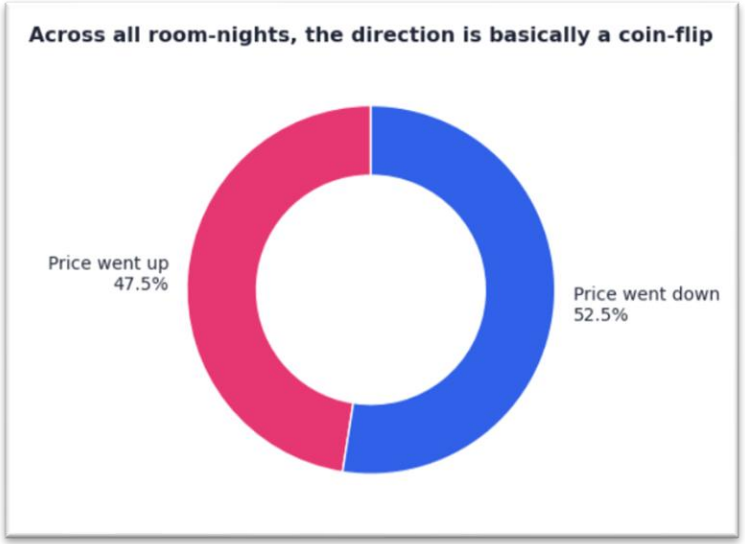
Model 4 : Same room-night, and the coin-flip

Compare the same hotel on the same night at different booking windows. The effect is statistically zero.



-$0.005 / day (p = 0.88)

95% CI [-0.06, +0.07] crosses zero : indistinguishable from no effect.



Why the average is zero

47.5% rise / 52.5% fall

For any given room-night the price is as likely to fall as to rise : a coin-flip. The rises and falls cancel out, which is why the model average lands at zero. Most slopes sit very close to zero, with only thin tails of dramatic movement.

It's selection bias : not a price trend

Premium inventory sells early; cheaper and distressed rooms remain near the date. That mix created the illusion.

“Prices rising”

A factual price claim. My analysis shows it is false on average and only true in narrow segments, so use it sparingly and only where proven.

47.5% ↑ / 52.5% ↓

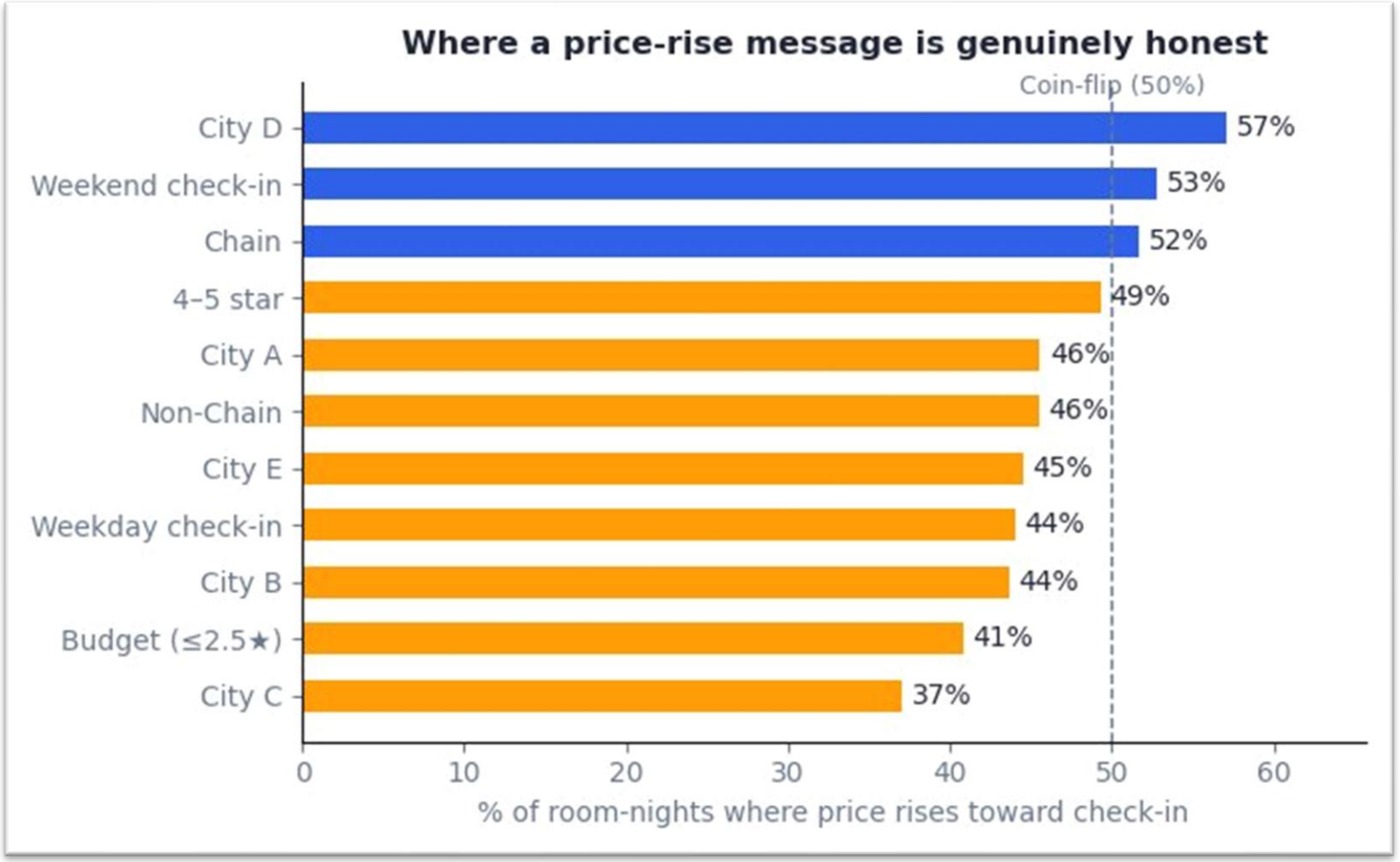
For any given room-night, price is as likely to fall as rise : a coin-flip.

Honest by need

A blanket “prices are rising” claim would mislead ~half of guests ; layer the price-rise message only on proven >50% segments.

The myth is debunked but the opportunity isn't dead. It just moves from whether prices rise to where they genuinely do.

Where the message is genuinely honest ?



HERO TARGET

City D + Weekend + 4-5 star

82%

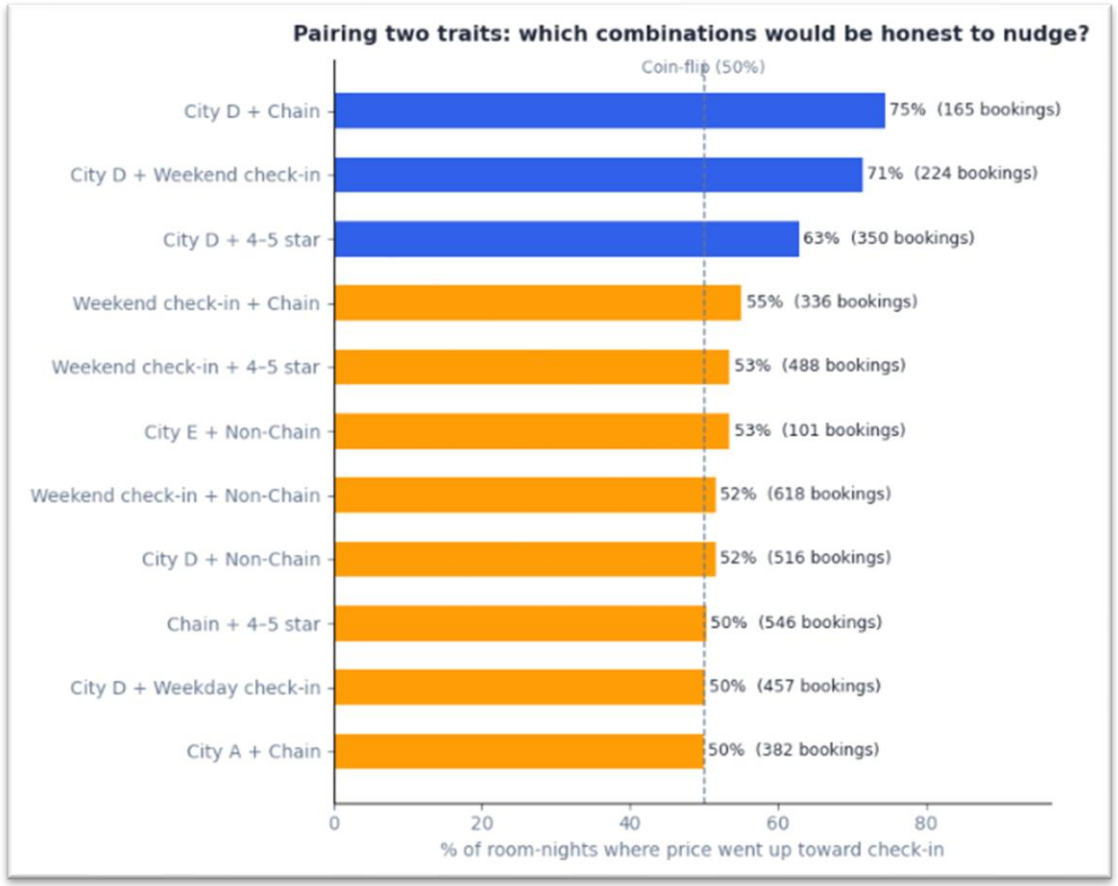
of room-nights genuinely rise toward check-in : here the message is accurate, not a guess.

Rule: price-rise wording only above the 50% line; target the top 3 segments & their combinations.

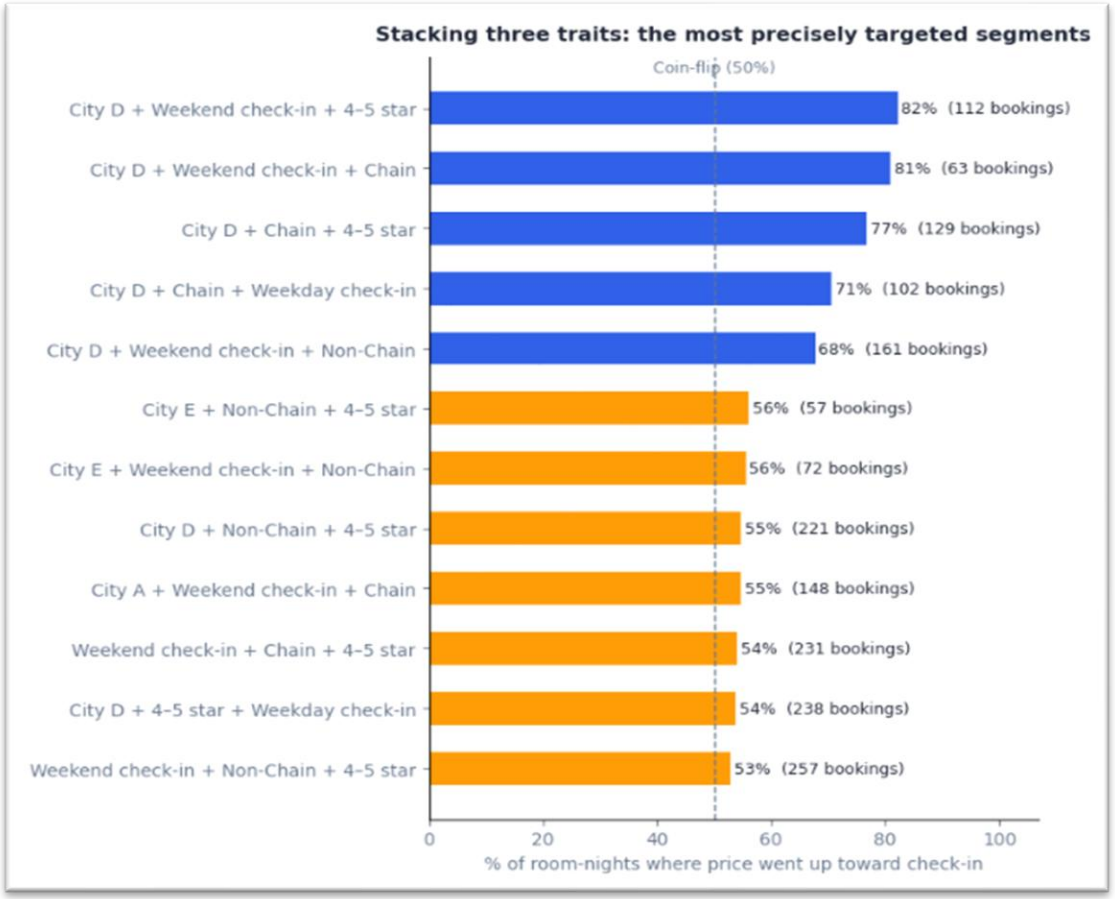
The average is flat : the segments are not

Stacking honest signals : the more aligned the segment, the more often prices genuinely rise.

Top 2-way combinations



Top 3-way combinations



Recommendations to the Product Owner

Targeted, honest, and proven before scaling.

Default to scarcity	Use availability-based prompts ("limited rooms left") that hold true regardless of price direction.
Reserve price-rise for proof	Show “prices may rise” only on City D + Weekend + 4-5 star (82%) and other >50% segments.
Place at high intent	Property page & booking form : closest to conversion, where a nudge converts without annoying browsers.
Keep it honest by design	Tie wording to live price & inventory signals : never a fake countdown. Protects trust and compliance.

Sizing the prize : and proving it safely

Illustrative, driven by conversion uplift on targeted segments.

Conservative

+0.5% conversion

≈ $1.3M

Base case

+1.0% conversion

≈ $2.5M

Optimistic

+2.0% conversion

≈ $5.1M

per 1M eligible sessions

avg booking value ≈ $252 (mean ADR $148 × LOS 1.7)

How I'd validate before scaling

A/B test

Property-page split: scarcity vs price-rise vs control. Primary metric = conversion.

Guardrails

Track cancellations, refunds, NPS. Auto-kill if trust metrics regress.

Truth audit

Fire the price-rise claim only where it proves ≥70% accurate.

Backtest

Re-check segment targeting on a fresh quarter before widening.

Assumptions, limitations & what's next

Stated up front : every conclusion rests on these, and each is testable.

Assumptions I made

- **Fri & Sat** are considered as weekend days due to price increase.
- ADR is a fair proxy for the price a shopper saw.
- Bookings imply the room was available at that price and time.
- Within a hotel+date, room-nights are comparable enough to compare.
- An urgency nudge can measurably shift conversion.

Limitations of the data

- Realised bookings only : no quoted prices or un-booked sessions.
- No room type, rate plan or cancellation policy in the data.
- No occupancy or competitor pricing : can't see demand pressure.
- One quarter, 5 cities, up to 3-night stays, 2016 vintage.

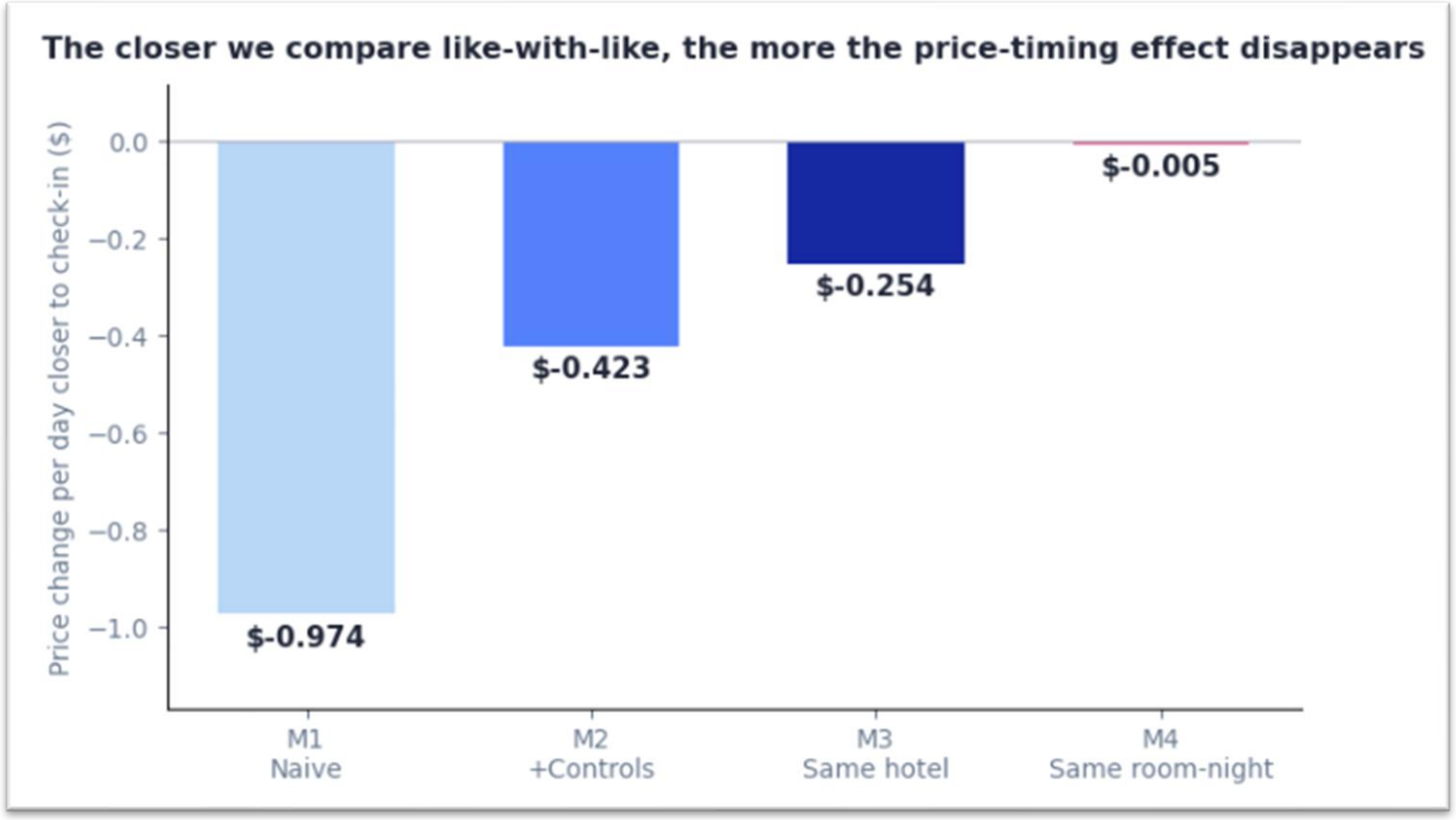
Data that would sharpen it

- Daily price quotes per room-night : a true price-over-time panel.
- Funnel events: impressions, clicks, conversion by step.
- Real-time occupancy / inventory remaining per property.
- Competitor rates and demand/event signals; customer segment.

Validating the assumptions: *the A/B test confirms the conversion assumption; a daily price-quote feed would directly test the ADR-as-signal and availability assumptions before any rollout.*

How I proved it : the statistical test

OLS with progressively tighter controls; the timing effect collapses to zero.



Read the collapse

Naive: -$0.97/day

+ Controls: -$0.42/day

Hotel fixed effects: -$0.25/day

Same room-night: -$0.005/day (p=0.88)

Comparing the same room-night to itself, the effect is statistically zero. The 39.5% was confounding : selection bias, not real price movement. Chain status adds ~$7 and is absorbed by hotel fixed effects.